Text Similarity Analysis-16/10/25(Thursday):

Report:

1. Project Goal

The primary objective of this project was to build an Extractive Text Summarization Model capable of generating concise summaries of long articles. The model was designed to identify and extract the most representative sentences, thereby preserving the article's core information and coherence.

2. Methodology: TextRank Algorithm

The project employed a feature-based approach using the TextRank algorithm, an unsupervised method based on the Google PageRank algorithm.

Step	Technique Applied	Description
Feature Extraction	TF-IDF (Term Frequency-Inverse Document Frequency)	Used to transform sentences into numerical vectors, emphasizing words unique and important to the document.
Sentence Scoring	Cosine Similarity & TextRank	The similarity between all sentence pairs was calculated using Cosine Similarity. These similarities formed a weighted graph where sentences are nodes. The TextRank algorithm then iterated on this graph to assign an importance score to each sentence.
Refinement	Positional Weighting	A small score bonus was added to the first sentence (index 0) to ensure the summary captured the article's initial topic statement, a crucial refinement step.

Summary Generation	Extraction and Ordering	The top N sentences (based on their TextRank score) were selected and re-ordered according to their original appearance in the article.

3. Implementation and Results

The model was developed in a Google Colab environment using Python libraries including NLTK (for tokenization), scikit-learn (for TF-IDF and similarity), and NetworkX (for the TextRank graph).

3.1 Data Used

For testing and demonstration, a single sample article on climate change was used:

- Original Article Length: 107 words (8 sentences).
- Target Summary Length (N): 3 sentences.

3.2 Sentence Scoring Outcome

The implementation of TextRank, coupled with positional weighting, successfully ranked the sentences. The top three scores and their corresponding indices were:

$$\begin{array}{c|c|c} \hline \textbf{Rank} & \textbf{Score} & \textbf{Original Index} \\ \hline 1 & 0.14 & 0 \\ 2 & 0.125 & 7 \\ 3 & 0.125 & 5 \\ \hline \end{array}$$

The positional boost ensured that the first sentence (Index 0) was prioritized, leading to a more topic-focused summary.

3.3 Final Extractive Summary

The model selected sentences corresponding to indices 0, 5, and 7 and combined them in chronological order to produce the final output:

The recent study on climate change revealed alarming trends. Transitioning to renewable energy sources like solar and wind is a critical step. Furthermore, biodiversity loss is accelerating, closely tied to these climatic shifts.

4. Conclusion and Achievements

The project successfully delivered a functional extractive summarization model based on the TextRank algorithm. The generated summary is concise and maintains high information density by capturing the article's core components: the main topic, a key solution, and a severe consequence.

Checklist Item	Project Achievement
Data/Preprocessing	Text was tokenized and vectorized using TF-IDF.
Scoring	TextRank was implemented, and the scoring was refined using positional weighting to ensure key introductory information was included.
Summary	A 3-sentence summary was generated, correctly ordered, and effectively preserved the article's core information.

This implementation serves as a strong foundation, and future work could involve experimenting with advanced feature representations, such as BERT embeddings, or implementing redundancy checks to further optimize the conciseness of the extracted summary.

Summary:

The project successfully implemented an Extractive Text Summarization model using the unsupervised TextRank algorithm.

Methodology

1. Text Representation: Sentences were first preprocessed and converted into numerical vectors using the TF-IDF (Term Frequency-Inverse Document Frequency) technique.

2. Sentence Similarity: The relationship between every pair of sentence vectors (V_A and V_B) was quantified using Cosine Similarity, which measures the angle between them:

$$\text{Cosine Similarity} (V_A, V_B) = \frac{V_A \cdot V_B}{\|V_A\| \cdot \|V_B\|}$$

3. Sentence Scoring: These similarity scores formed a weighted network graph, where sentences are nodes and similarity scores are edges. The TextRank algorithm (based on PageRank) was applied to this graph to iteratively calculate the final importance score of each sentence.
4. Refinement: To improve summary quality, positional weighting was applied to boost the score of the initial sentence, ensuring the topic was captured.
5. Summary Generation: The top $N=3$ highest-scoring sentences were extracted and presented in their original document order to form the final, coherent summary.

Final Result

The model generated a 3-sentence summary that successfully captures the main topic, a key solution, and a major consequence of the input article.

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$$\{\textbf{cosine similarity}(\mathbf{Va},\mathbf{Vb})=\frac{\mathbf{Va.Vb}}{||\mathbf{Va}||.||\mathbf{Vb}||}$$

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